SRI SATHYA SAI INSTITUTE OF HIGHER LEARNING

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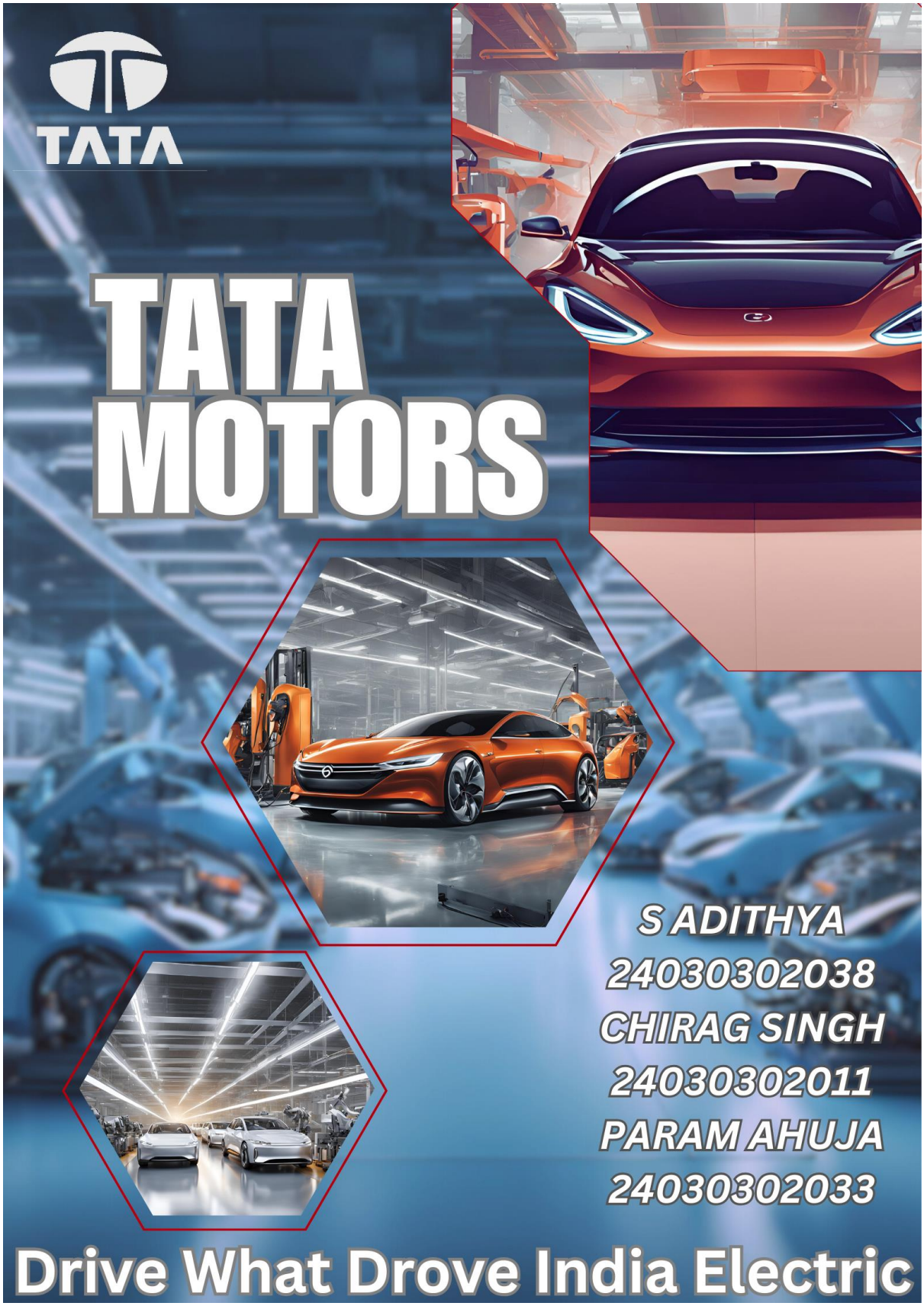
MBAG – 105

ACCOUNTING FOR FINANCIAL DECISIONS

ASSIGNMENT

FINANCIAL STATEMENT ANALYSIS OF TATA MOTORS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INTRODUCTION

Tata Motors, part of the Tata Group is a USD 44 billion organisation, leading global automobile manufacturers of cars, utility vehicles, pickups, trucks and buses. Tata Motors is India's market leader when it comes to the commercial vehicle segment and is among the top 3 in the passenger vehicles market. It's cutting-edge designs on a quest to excel and its promise to deliver superior and satisfactory values to its customers helps it to stay ahead in each curve.

Tata Motors was founded in 1945, first as a locomotive manufacturer. After a joint venture with Mercedes Benz, Tata Group entered into the commercial vehicle sector in 1954. Its first manufacturing facility was set up in Jamshedpur. From there on Tata Motors had multiple milestones in its history, each milestone putting them multiple steps ahead of its competitors. In 2008, it acquired the Jaguar & Land Rover for $2.3 billion, when Ford was facing bankruptcy during the 2008 Great Recession.

Here is a brief snippet of their operations from their Annual Report for 2023-2024.

TATA MOTORS Vision & Mission

Sustainability Highlights

Our key presence

Across
~125
Countries

25
Manufacturing facilities (consolidated)

9
R&D sites

>9,400
Touchpoints (consolidated)

>91,000
Collective workforce strength

India (1)

9,51,098
Vehicles sold

₹1,27,755 crore
Revenue

15
Manufacturing sites

3
R&D sites

North America (2)

1,05,623
Vehicles sold

₹67,496 crore
Revenue

1
R&D site

China (3)

52,553
Vehicles sold (excl. CJLR)

₹57,184 crore
Revenue

1
Joint manufacturing site

Europe (4)

80,837
Vehicles sold

₹55,528 crore
Revenue

2
Manufacturing sites

1
R&D site

UK (5)

82,375
Vehicles sold

₹55,009 crore
Revenue

5
Manufacturing sites

3
R&D sites

TATA MOTOR's Global Presence

FINANCIAL STATEMENT ANALYSIS

OBJECTIVE

The objective is to decide whether to invest in Tata Motors or not by examining and understanding the company's income statement, balance sheet and cash flow statement for the last 5 years and interpreting it.

PRELIMINARY DATA COLLECTION

For analysis, consolidated financial statements are taken. This is because, at the time of investing, a company's growth prospects are affected by what its subsidiaries are earning (in other words, the subsidiary's financial figures have either a direct or indirect impact on the parent company's financials). Therefore, to arrive at a better conclusion. Consolidated financial statements are taken.

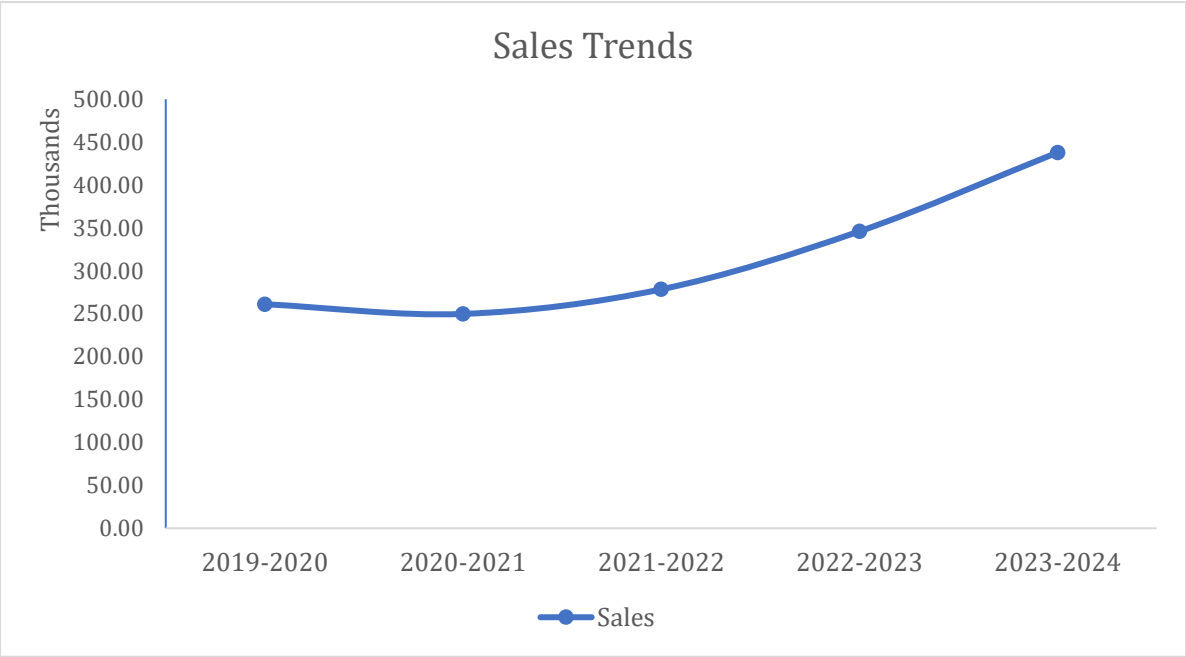
Note: *For calculations, most of the figures have been taken from Screener. In, unless specified in the footnotes of the respective ratio calculations.*

For better understanding, the reference links for the financial statements used for analysis have been provided on the reference page. Therefore, only the figures required for calculation are used here)

INCOME STATEMENT & BALANCE SHEET ANALYSIS

Narration[1]	2019-2020	2020-2021	2021-2022	2022-2023	2023-2024
Sales	**2,61,067.97**	**2,49,794.75**	**2,78,453.62**	**3,45,966.97**	**4,37,927.77**
Expenses	2,43,080.90	2,17,507.32	2,53,733.53	3,14,151.17	3,78,389.43
Operating Profit	17,987.07	32,287.43	24,720.09	31,815.80	59,538.34
Other Income	101.71	11,117.83	2,424.05	6,663.97	5,672.66
Depreciation	21,425.43	23,546.71	24,835.69	24,860.36	27,270.13
Interest	7,243.33	8,097.17	9,311.86	10,225.48	9,985.76
Profit before tax	**10,579.98**	**10,474.28**	**7,003.41**	**3,393.93**	**27,955.11**
Tax	395.25	2,541.86	4,231.29	704.06	3,851.64
Net profit	**12,070.85**	**13,451.39**	**11,441.47**	**2,414.29**	**31,399.09**

[1] Negative Values are highlighted in Red

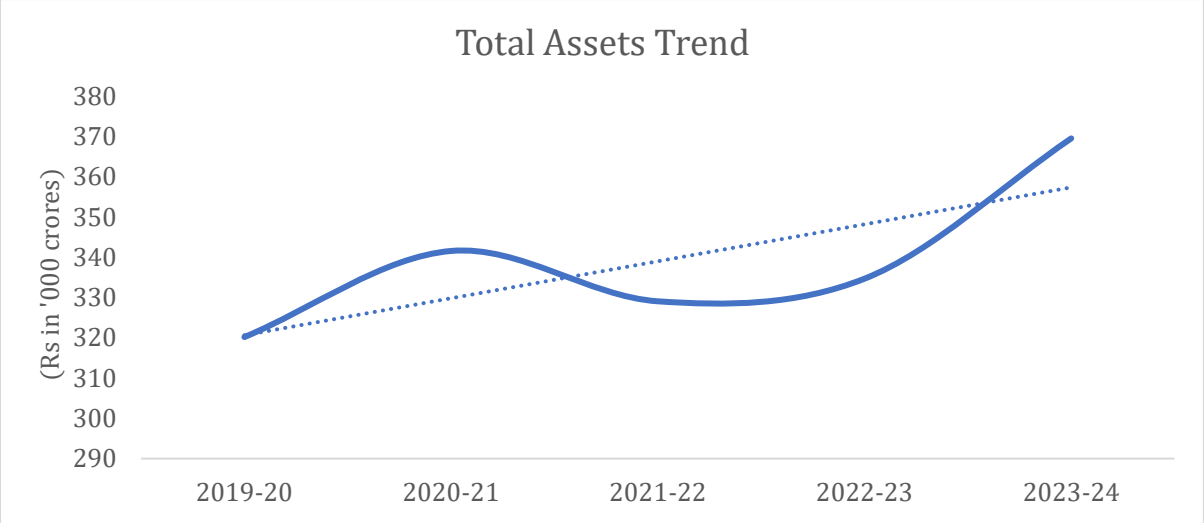


The sales of the company have been on the rise slowly. This indicates better strategic decisions to generate more revenue for the company. Also, due to an additional subsidiary acquired by Tata Motors in January 2023, it added to the overall sales value of Tata Motors.

Here is a summarized Balance Sheet for the past 5 years: -

Narration	2019-2020	2020-2021	2021-2022	2022-2023	2023-2024
Equity Share Capital	719.54	765.81	765.88	766.02	766.50
Reserves	61,491.49	54,480.91	43,795.36	44,555.77	84,151.52
Borrowings	1,24,787.64	1,42,130.57	1,46,449.03	1,34,113.44	1,07,262.50
Other Liabilities	1,33,180.72	1,44,192.62	1,38,051.22	1,55,239.20	1,77,340.09
Total	**3,20,179.39**	**3,41,569.91**	**3,29,061.49**	**3,34,674.43**	**3,69,520.61**
Net Block	1,27,107.14	1,38,707.61	1,38,855.45	1,32,079.76	1,21,285.46
Capital Work in Progress	35,622.29	20,963.93	10,251.09	14,274.50	35,698.43
Investments	16,308.48	24,620.28	29,379.53	26,379.16	22,971.07
Other Assets	1,41,141.48	1,57,278.09	1,50,575.42	1,61,941.01	1,89,565.65
Total	**3,20,179.39**	**3,41,569.91**	**3,29,061.49**	**3,34,674.43**	**3,69,520.61**

The graph below shows the growth trend of Total Assets of Tata Motors over the past 5 years. It has been gradually increasing every year, indicating a strategic investment into PPEs to foster growth for the future.



RATIO ANALYSIS

Liquidity Ratios are calculated to indicate the short-term repayment capacity of the firm. Majorly, two ratios are looked at when a company's liquidity ratios are calculated. These are:

Current Ratio & Quick Ratio

Current Ratio[2]: $\frac{\textit{Current Assets}}{\textit{Current Liabilites}}$

Components	2019-2020	2020-2021	2021-2022	2022-2023	2023-2024
Current Assets	1,91,392.82	1,46,666.84	1,46,453.69	1,50,700.69	1,67,718.24
Current Liabilities	1,40,454.05	1,57,749.18	1,50,679.69	1,57,027.33	1,73,617.00
Ratio	1.36	0.93	0.97	0.96	0.97

(Amount in Rs. Crore)

Current Ratio indicates a company's ability to meet its short-term debts, i.e., how much of its current liabilities are met by current assets, without taking additional loans.

Quick Ratio: $\frac{\textit{Current Assets} - \textit{Inventories}}{\textit{Current Liabilites}}$

Components	2019-2020	2020-2021	2021-2022	2022-2023	2023-2024
Quick Assets	1,53,935.94	1,10,578.25	1,11,213.35	1,09,945.30	1,19,929.95
Current Liabilities	1,40,454.05	1,57,749.18	1,50,679.69	1,57,027.33	1,73,617.00
Ratio	1.10	0.70	0.74	0.70	0.69

(Amount in Rs. Crore)

Quick Ratio indicates a company's ability to meet its immediate liabilities, without having to convert its stock.

Analysis:

It could be seen that the current ratio has been decreasing. Therefore, it indicates that the company's liquidity position is weakening slightly. However, Tata Motors has managed to

[2] The figures have been taken from the Annual Report of Tata Motors.

keep it at a constant level. Therefore, compared to the industry ratio of 2:00; Tata Motors is looking good in terms of its performance.

Looking at a company's solvency, majorly from an investor viewpoint, two ratios are calculated:

Debt-Equity Ratio & Interest Coverage Ratio.

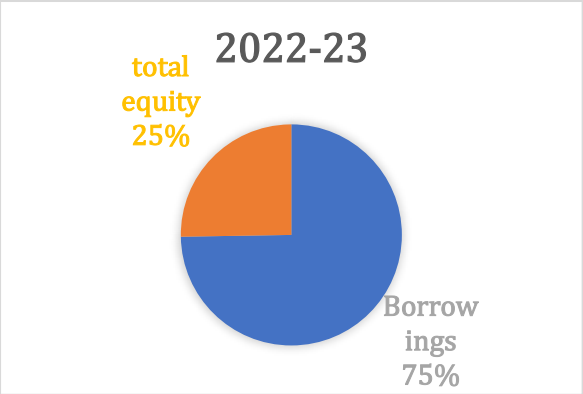
Shown below is the calculation for the 2 ratios over the past 5 years:

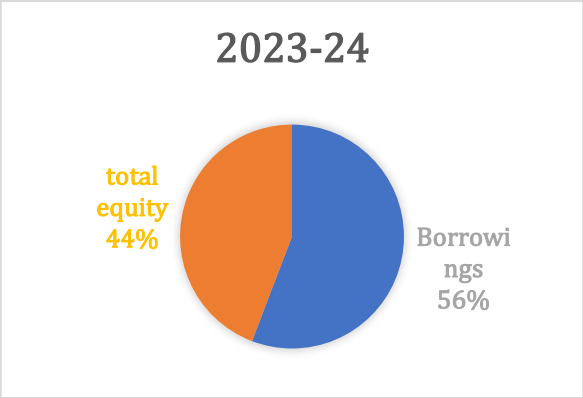
Debt-Equity Ratio: $\frac{\textit{Total Debt (Borrownigs+Lease Liabilities)}}{\textit{Total Equity (attributable to shareholders)}}$

Components	2019-2020	2020-2021	2021-2022	2022-2023	2023-2024
Total Debt[3]	1,24,787.64	1,42,130.57	1,46,449.03	1,34,113.44	1,07,262.50
Equity[4]	62,211.03	55,246.72	44,561.24	45,321.79	84,918.02
Ratio	2.01	2.57	3.29	2.96	1.26

(Amount in Rs. Crore)

This ratio indicates the relative portion of debt and equity that is used to finance a company's assets. It shows the company's financial leverage. High Ratio indicates high financial risk.





As can be seen, from 2022-2023, the debt content (represented by the blue pie) has reduced to 56% of the capital structure in 2023 – 2024 from 75% of the capital structure.

Interest Coverage Ratio: $\frac{\textit{Profit before Tax+Finacne Cost}}{\textit{Total Interest (Finance Cost)}}$

Components	2019-2020	2020-2021	2021-2022	2022-2023	2023-2024
Profit before Tax	10,579.98	10,474.28	7,003.41	3,393.93	27,270.13
Finance Cost	7,243.33	8,097.17	9,311.86	10,225.48	9,985.76
EBIT	3,336.65	2,377.11	2,308.45	13,619.41	37,255.89
Ratio	0.46	0.29	0.25	1.33	3.73

(Amount in Rs. Crore)

[3] Total Debt consists of Borrowings + Lease Liabilities. Both Non-Current and Current liability of the Non-Current Debt are considered.
[4] "Equity" does include that which is attributable to minority stakeholders as they do not own shares in the company.
NEGATIVE FIGURES ARE HIGHLIGHTED IN RED (FOR ALL CALCULATIONS)

This ratio indicates how many times a company will be able to meet its interest liability is covered by its profits. A high ratio usually indicates that a larger proportion of its interest can be met out of current earnings.

The ratio is seen to be increasing because of as mentioned earlier, the focus on repaying the debt to increase the equity part of the capital structure.

Analysis:

It could be seen that the DE ratio is decreasing, whereas the ICR is increasing. This indicates that the company will not face any trouble in meeting its interest obligations because of decreasing debt and increasing EBIT. Therefore, this is a good sign showcasing that it will face any solvency issues in the near future.

A company's performance from the point of its efficient use of resources is majorly analysed by finding the Debt Collection Period, Inventory Turnover Period, and Asset Turnover Ratio.

Shown below are the calculations for these ratios:

$$\text{Inventory Turnover Ratio: } \frac{\textit{Total Revenue}}{\textit{Average Inventory}}$$

Components	2019-2020	2020-2021	2021-2022	2022-2023	2023-2024
Inventory	37,456.88	36,088.59	35,240.34	40,755.39	47,788.29
COGS[5]	1,92,621.07	1,82,360.88	2,06,761.79	2,46,275.56	3,04,165.49
Average Inventory	38,235.31	36,772.74	35,664.47	37,997.87	44,271.84
Ratio (times)	5.04	4.96	5.80	6.48	6.87
Inventory Turnover Days	72.45	73.60	62.96	56.32	53.13

(Amount in Rs. Crore)

This ratio indicates how quickly Tata Motors can convert its inventory into finished goods thereby converting it into sales. A low period indicates that it is efficient in using its resources to convert its inventory. It could be seen that this ratio is decreasing year on year, indicating better usage of its resources to enable quick sales.

$$\text{Asset Turnover Ratio: } \frac{\textit{Total Revenue}}{\textit{Total Assets}}$$

Components	2019-2020	2020-2021	2021-2022	2022-2023	2023-2024
Revenue	2,61,067.97	2,49,794.75	2,78,453.62	3,45,966.97	4,37,927.77
Total Assets[6]	3,22,121.26	3,43,125.80	3,30,619.93	3,36,081.38	3,70,663.96
Ratio	0.81	0.73	0.84	1.03	1.18

(Amount in Rs. Crore)

[5] The Figures have been taken from the Annual Report of Tata Motors.
[6] The values for Total Assets have been taken from the Annual Report of Tata Motors

This ratio indicates how efficiently Tata Motors can use its assets to generate revenue. It could be seen that Tata Motors has somewhat of a consistent performance in this area, indicating a constant yet comparatively higher efficiency.

Trade Receivable Turnover Ratio: $\frac{\textit{Total Revenue}}{\textit{Average Trade Receivables}}$

Components	2019-2020	2020-2021	2021-2022	2022-2023	2023-2024
Trade Receivables	11,172.69	12,679.08	12,442.12	15,737.97	16,951.81
Revenue	2,61,067.97	2,49,794.75	2,78,453.62	3,45,966.97	4,37,927.77
Average Trade Receivables	15,084.43	11,925.89	12,560.60	14,090.05	16,344.89
Ratio (Times)	17.31	20.95	22.17	24.55	26.79
Debt Collection Period (Days)	21.08	17.43	16.46	14.87	13.62

(Amount in Rs. Crore)

This ratio indicates how strict Tata Motors' debt collection policies are. A lower collection period indicates a strict credit policy. It could be seen that the collection period is decreasing gradually, indicating that the company is putting up a strong credit policy to recover its debts, indicating enough cash availability with the firm to meet its current obligations.

Looking at a company's profitability and deciding whether there is scope for investors to invest, majorly, Net Profit Margin, Return on Equity, Return on Assets, Earnings Per Share & Price-Earnings Ratio are looked at. In some cases, even the Price Book Ratio is also looked at.

Shown below are the above-mentioned ratios for the past 5 years and their analysis:

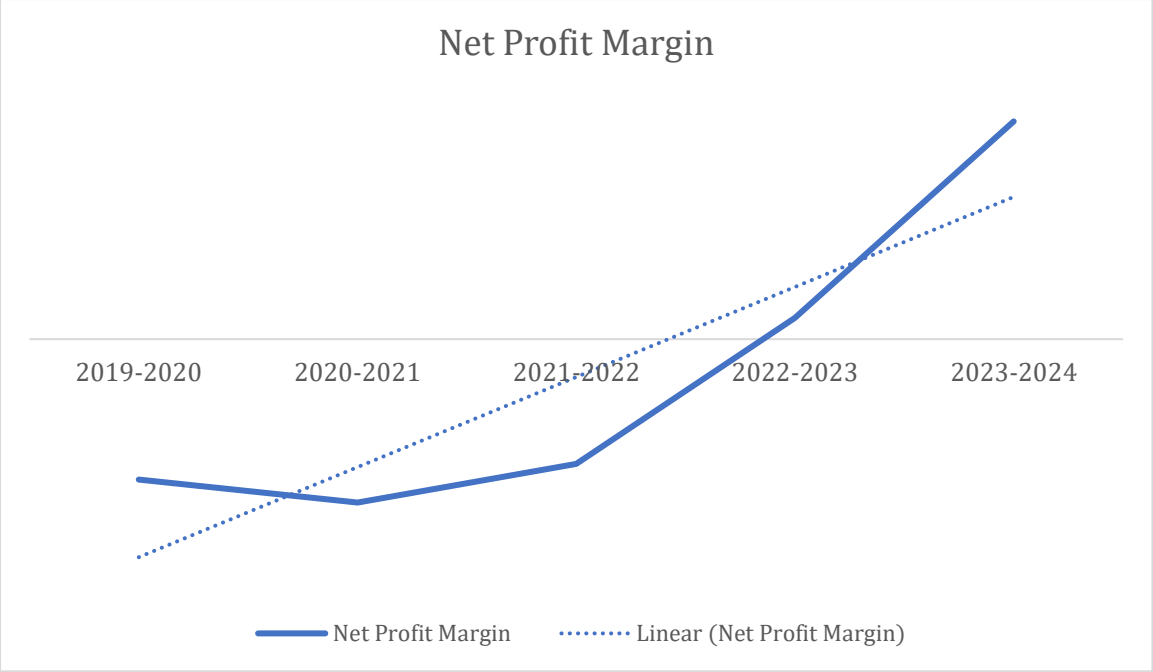
Net Profit Margin: $\frac{\textit{Net Profit after tax(attributable to shareholders)}}{\textit{Total Revenue}}$

Components	2019-2020	2020-2021	2021-2022	2022-2023	2023-2024
Net Profit after tax[7]	12,070.85	13,451.39	11,441.47	2,414.29	31,399.09
Revenue	2,61,067.97	2,49,794.75	2,78,453.62	3,45,966.97	4,37,927.77
Net Profit Margin	4.62	5.38	4.11	0.70	7.17

(Amount in Rs. Crore)

This ratio indicates how much is the company able to retain its revenue as profits. A high ratio indicates a good performance.

[7] The values do not include the amount that is attributable to the minority shareholders.



Tata Motors, as can be seen, has ensured to increase its Net Margin from negative values to positive values. This indicates that it has aimed to achieve this measure by initiating some cost-cutting measures to ensure its overall profitability rises. It has increased its net profit margin from negative figures to positive values in just 3 years, indicating a good sign for the investors. The dotted blue line (trendline), indicates possible increasing net profit margin in the coming years.

Return on Equity (%): $\frac{\textit{Net Profit after tax(attributable to shareholders)}}{\textit{Net Worth (Equity attributable to shareholders)}}$

Components	2019-2020	2020-2021	2021-2022	2022-2023	2023-2024
Net Profit after Tax	12,070.85	13,451.39	11,441.47	2,414.29	31,399.09
Net Worth (Equity)	62,211.03	55,246.72	44,561.24	45,321.79	84,918.02
Return on Equity	19.40	24.35	25.68	5.33	36.98

(Amount in Rs. Crore)

This ratio shows the return generated to shareholders out of the amount that they invested into the business. A high ratio indicates a good performance. It is seen that the ratio has taken a great leap from what it was in 2019-2020. This indicates that the company is earning more per rupee invested by shareholders.

Return on Assets (%): $\frac{\textit{Net Profit after tax(attributable to shareholders)}}{\textit{Total Assets}}$

Components	2019-2020	2020-2021	2021-2022	2022-2023	2023-2024
Net Profit after Tax	12,070.85	13,451.39	11,441.47	2,414.29	31,399.09
Total Assets	3,22,121.26	3,43,125.80	3,30,619.93	3,36,081.38	3,70,663.96
Return on Assets	3.75	3.92	3.46	0.72	8.47

(Amount in Rs. Crore)

This ratio indicates what percentage of assets are contributing to earning a profit for the firm. The higher the ratio, it indicates a greater portion of assets is utilized to generate income for the firm. From the calculation, it could be seen that the company is doing better to generate more profits out of its usage of assets, showcasing a good sign to investors.

Earnings Per Share: $\frac{\textit{Net Profit after tax(attributable to shareholders)}}{\textit{Total No of Shares (Equity)}}$

Components	2019-2020	2020-2021	2021-2022	2022-2023	2023-2024
Basic EPS	39.08	40.51	34.45	7.27	94.47

(Amount in Rs. Crore)

When it comes to investment decisions, this ratio is the most looked after. It shows the earnings that shareholders are getting for each rupee invested. An EPS of 94.47 Rs/ Share indicates a very good performance. It indicates that if a shareholder has invested Rs. 100 in Tata Motors, his earnings are (100*94.47) Rs. 9,447.

P/E Ratio **(times):** $\frac{\textit{Market Price of Share}}{\textit{Earnings Per Share}}$

Components	2019-2020	2020-2021	2021-2022	2022-2023	2023-2024
P/E Ratio	1.82	7.45	12.59	57.89	10.51

(Amount in Rs. Crore)

This ratio shows how much an investor is willing to invest in a company for Rs. 1 earned by the company. A high ratio indicates that the company can expect growth. It could be seen that for the years 2023-2024, the P/E Ratio has declined from 57.89 to 10.51 from 2022-2023, indicating that the company's stocks are either undervalued or it is at risk of fewer investment opportunities. However, it is also to be seen that the ratio has not dropped below the least of the past 4 years (1.82 times). This indicates a good performance by the company.

CASH FLOW STATEMENT ANALYSIS

Cash Flow Statement Analysis provides insights into how a company has generated and utilized cash during a particular year. In other words, it shows the sources from where it is collecting its cash and the avenues into which that cash is invested.

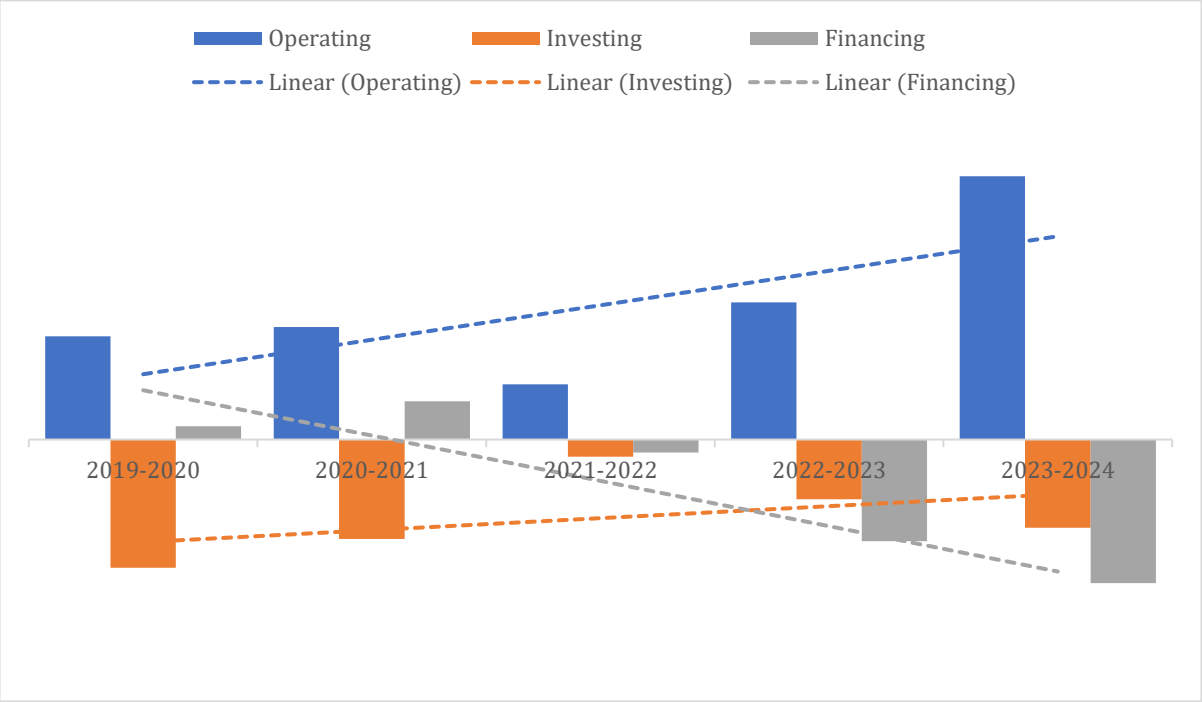
Shown below is an extract of the final cash flow from each activity – Operating, Investing and Financing activities.

Narration	2019-2020	2020-2021	2021-2022	2022-2023	2023-2024
Cash from Operating Activity	26,632.94	29,000.51	14,282.83	35,388.01	67,915.36
Cash from Investing Activity	-33,114.55	-25,672.50	-4,443.66	-15,417.17	-22,781.56
Cash from Financing Activity	3,389.61	9,904.20	-3,380.17	-26,242.90	-37,005.99
Net Cash Flow	**-3,092.00**	**13,232.21**	**6,459.00**	**-6,272.06**	**8,127.81**

For the year 2023-2024, the company has leap jumped from negative 6,272.06 (cash outflow) crores to positive 8,127.81 crores. Looking at the Cash Flow from operating activities, it is seen that there is a 100% change in the cash inflow. This is because of better debt collection measures, reduced finance costs, better usage of raw materials to convert to finished goods (which was also evident from Trade Receivables Turnover Ratios, Inventory Turnover Ratios & Debt – Equity Ratios)

Looking at Cash Flow from Investing Activities shows that Tata Motors' investments in PPE have increased by close to 4500 crores from 2023 YE to 2024 YE, indicating better scope for future growth and expansion, particularly with the rising EV market.

Looking at Financing Activities, it was found that due to lower dependence on debt, there is a larger part of outflow in the form of debt repayment.



FINAL ANALYSES:

The figures point out a good performance by Tata Motors and its subsidiaries. This is indicative of the fact that the company is on the path towards further expansion and growth.

The graph above shows the trend of various activities concerning cash flow. It can be seen that Financing activities have been decreasing, indicating lower reliance on external financing (through debt) and higher dividend payments to shareholders (a positive sign to investors). Increasing Cash flow from operating activities indicates a positive and strong focus on core business operations. Cash flow from investing activities has been on the rise. These marks show a good sign for investors.

TREND ANALYSIS – STOCK PRICE

The above picture shows the movement of the stock price of Tata Motors from November 2021 to October 2024. The smooth increasing curved line is the 200-day moving average. The Moving Average line helps an investor to predict the likely movement of stock price (upwards or downwards). When the prices are constantly trading above the 200-day moving average, it generally indicates an increasing price of the stock in the future. When it is below, it is the converse. From the graph picture, it can be seen that the shares are traded at prices above the moving average line, thus, the prices were on the rise.

If we look at the values for the month of October 2024, it is below the 200-day moving average line, indicating a slight reduction in price in the future. However, due to the company's strong focus on increasing value to its shareholders, it is expected that there won't be much drop in prices.

MISCELLANEOUS ANALYSIS

Some factors that affect an investment decision are analysing what prospects the chairman is having towards Tata Motors and some external factors that affect the company's performance. For instance, the chairman *N. Chandrasekaran* has reiterated that the company is going for a strong growth opportunity given the increasing EV sector. The company is expected to focus on increasing its revenue growth, strong free cash flows etc.

Also, the recent unfortunate event of Mr Ratan Tata's demise has affected the share price. Events like these have a lot of impact on the share prices, as share prices are majorly based on investor sentiment.

However, it is also hoped that the share price will not take a sharp dip. This is because based on a report from India Today, Noel Tata has been successful in growing Tata international revenue from 4,200 crores to 25,220 crores between August 2010 and

November 2011. Thus, with a prominent person occupying the chair of the Tata Group. It is expected that the share price will increase though for the short term, it will decrease.

CONCLUSION

Based on the analysis it seems that the company's financials are showing a strong sign to the investors. The exceptional performance of Tata Motors in the automobile sector, it is showing strong confidence to the investors that the company will perform better in the future.

In this regard, it is advised to invest in Tata Motors. With the falling prices, it will be better for an investor if (s)he invests now, else the price is expected to increase in the future. Therefore, it is concluded that investing in Tata Motors is a good investment option.

REFERENCES

https://www.screener.in/company/TATAMOTORS/consolidated/

https://www.tatamotors.com/

https://en.wikipedia.org/wiki/Tata_Motors

https://www.moneycontrol.com/india/stockpricequote/auto-lcvshcvs/tatamotors/TM03